Grad-CAM examples (MobileNetV2, full fine-tuning, seed 0)

10 correct predictions (one per class, first in test-file order)

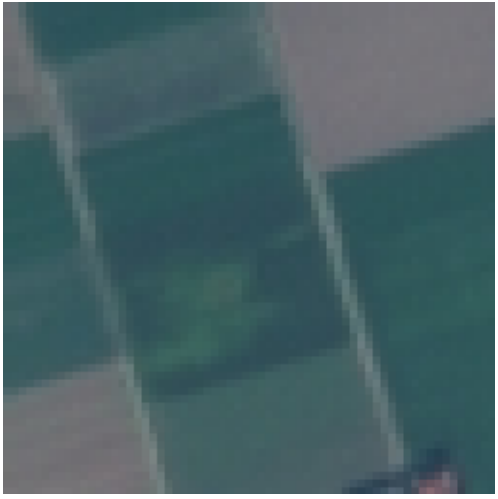
10 wrong predictions: one from each of the 10 most frequent confusion pairs

Each page: original tile | Grad-CAM heat map | heat map over the tile

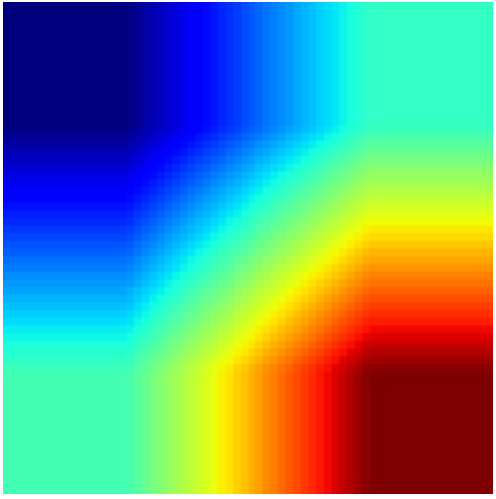
Red = where the model looked most. Blue = least. Source: src/gradcam.py.

1. CORRECT | actual: AnnualCrop | predicted: AnnualCrop (99.9% confident)
eurosat/2750/AnnualCrop/AnnualCrop_1002.jpg

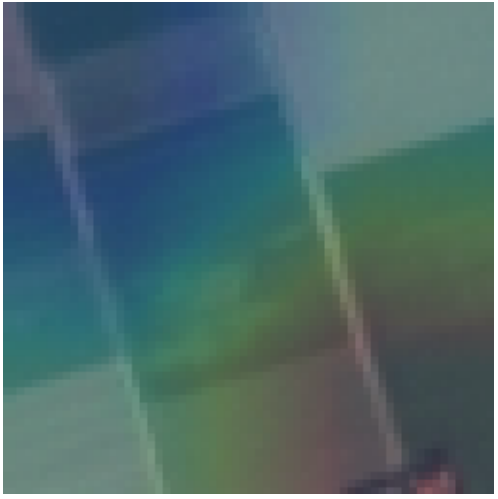
Original tile



Grad-CAM heat map

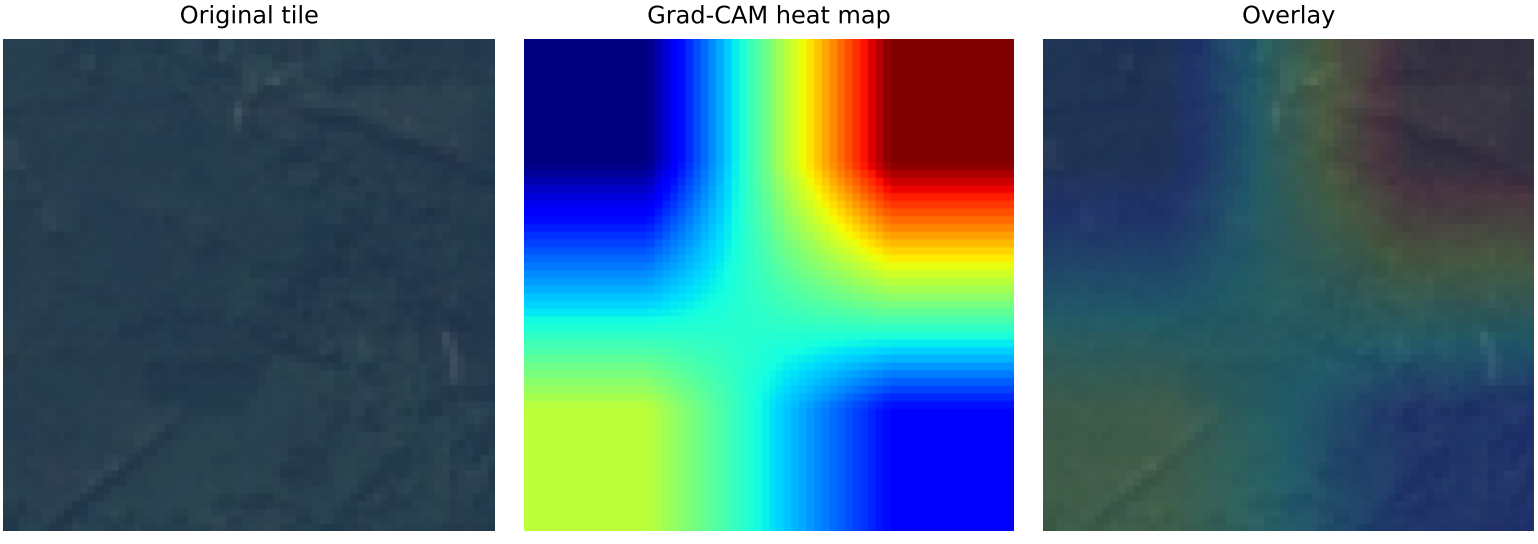


Overlay



Heat map: where the model looked. Use with care: see report section 6 (control experiment).

2. CORRECT | actual: Forest | predicted: Forest (100.0% confident)
eurosat/2750/Forest/Forest_1.jpg



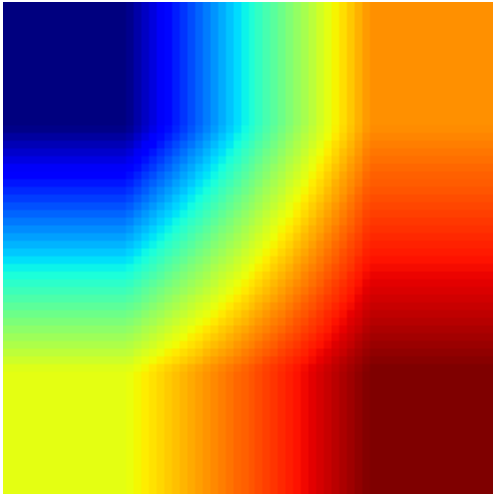
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

3. CORRECT | actual: HerbaceousVegetation | predicted: HerbaceousVegetation (99.9% confident)
eurosat/2750/HerbaceousVegetation/HerbaceousVegetation_10.jpg

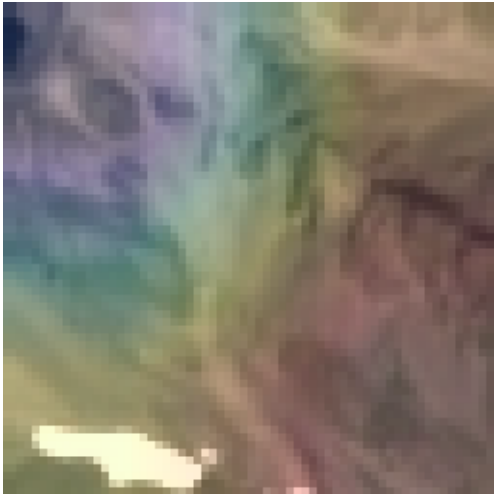
Original tile



Grad-CAM heat map



Overlay



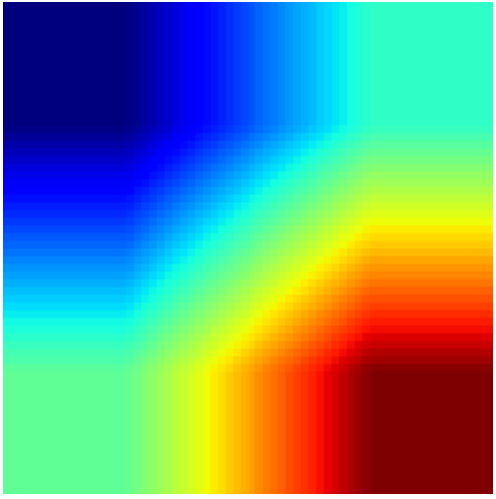
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

4. CORRECT | actual: Highway | predicted: Highway (99.9% confident)
eurosat/2750/Highway/Highway_1006.jpg

Original tile



Grad-CAM heat map



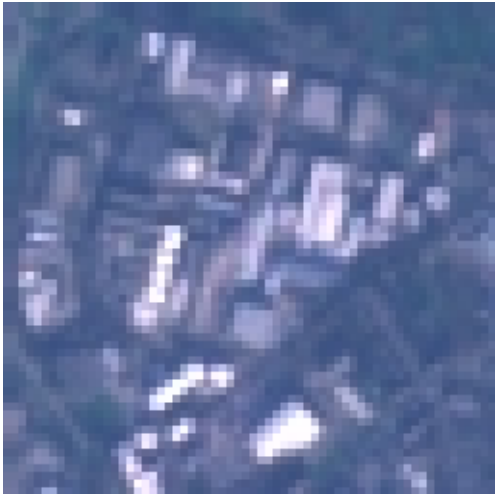
Overlay



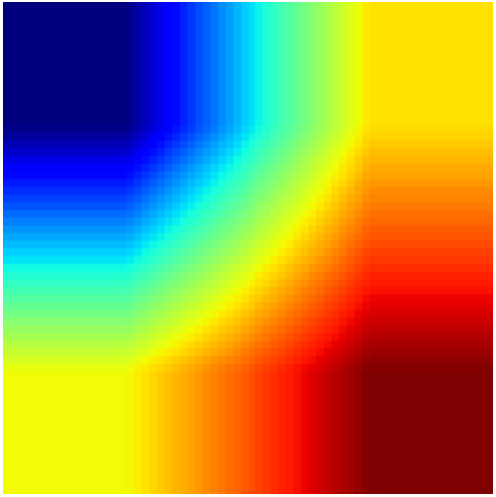
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

5. CORRECT | actual: Industrial | predicted: Industrial (100.0% confident)
eurosat/2750/Industrial/Industrial_1009.jpg

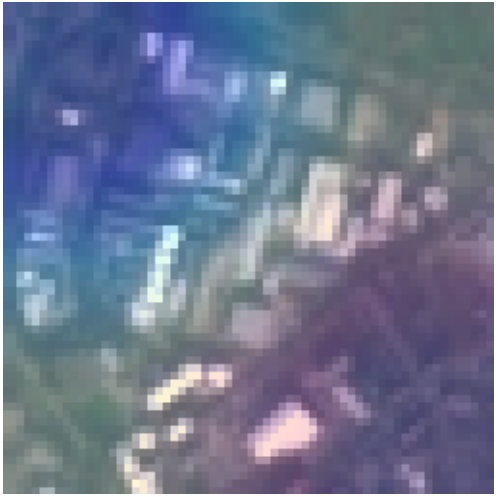
Original tile



Grad-CAM heat map



Overlay



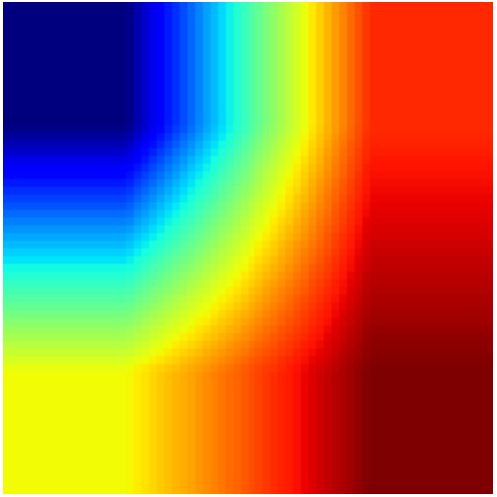
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

6. CORRECT | actual: Pasture | predicted: Pasture (100.0% confident)
eurosat/2750/Pasture/Pasture_1008.jpg

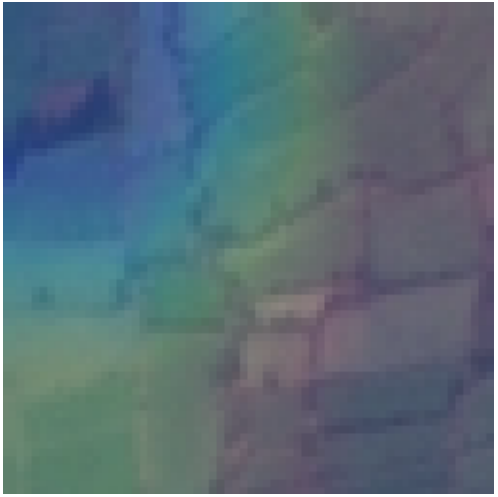
Original tile



Grad-CAM heat map



Overlay



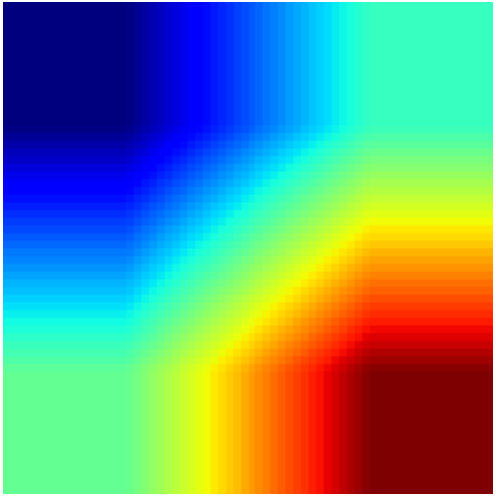
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

7. CORRECT | actual: PermanentCrop | predicted: PermanentCrop (99.2% confident)
eurosat/2750/PermanentCrop/PermanentCrop_1009.jpg

Original tile



Grad-CAM heat map



Overlay



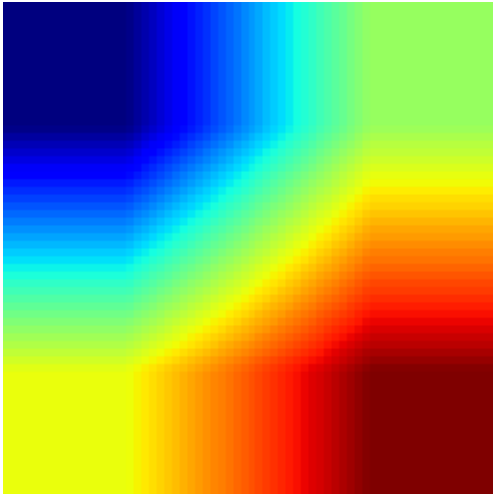
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

8. CORRECT | actual: Residential | predicted: Residential (100.0% confident)
eurosat/2750/Residential/Residential_1016.jpg

Original tile



Grad-CAM heat map



Overlay



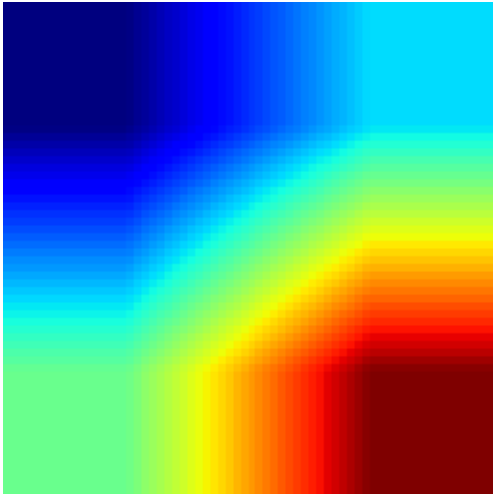
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

9. CORRECT | actual: River | predicted: River (96.6% confident)
eurosat/2750/River/River_100.jpg

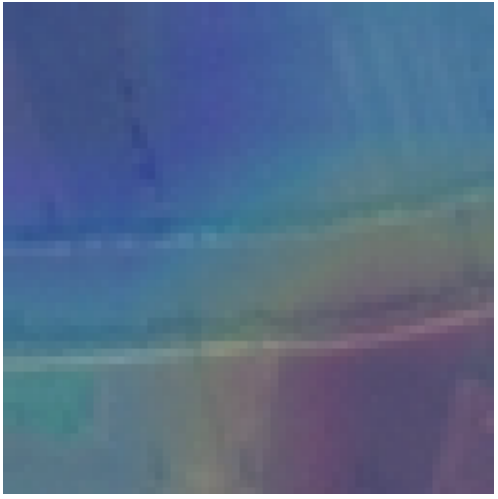
Original tile



Grad-CAM heat map



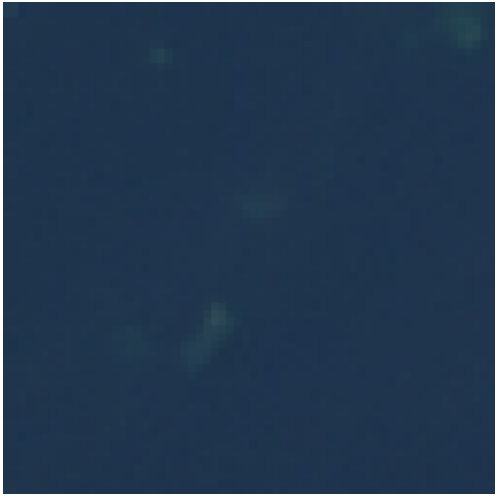
Overlay



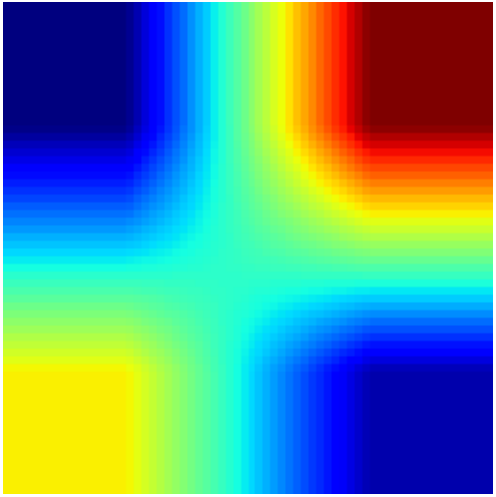
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

10. CORRECT | actual: SeaLake | predicted: SeaLake (100.0% confident)
eurosat/2750/SeaLake/SeaLake_10.jpg

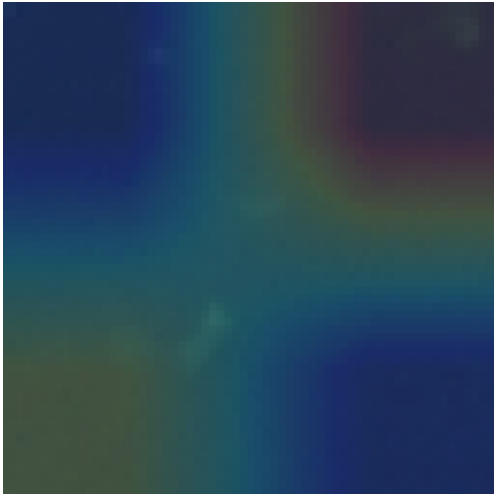
Original tile



Grad-CAM heat map



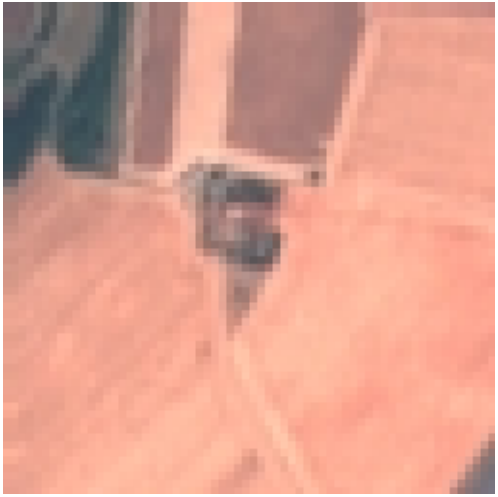
Overlay



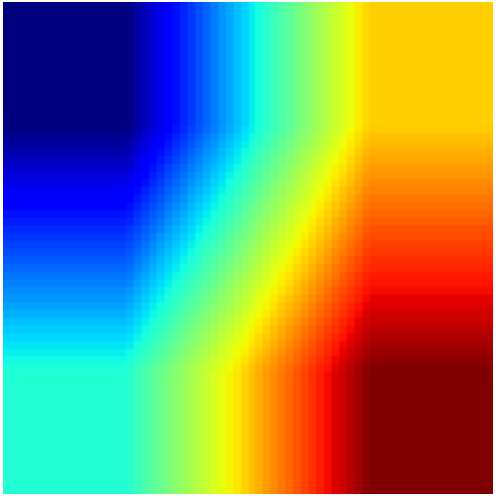
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

11. WRONG | actual: AnnualCrop | predicted: PermanentCrop (61.3% confident)
eurosat/2750/AnnualCrop/AnnualCrop_1094.jpg

Original tile



Grad-CAM heat map

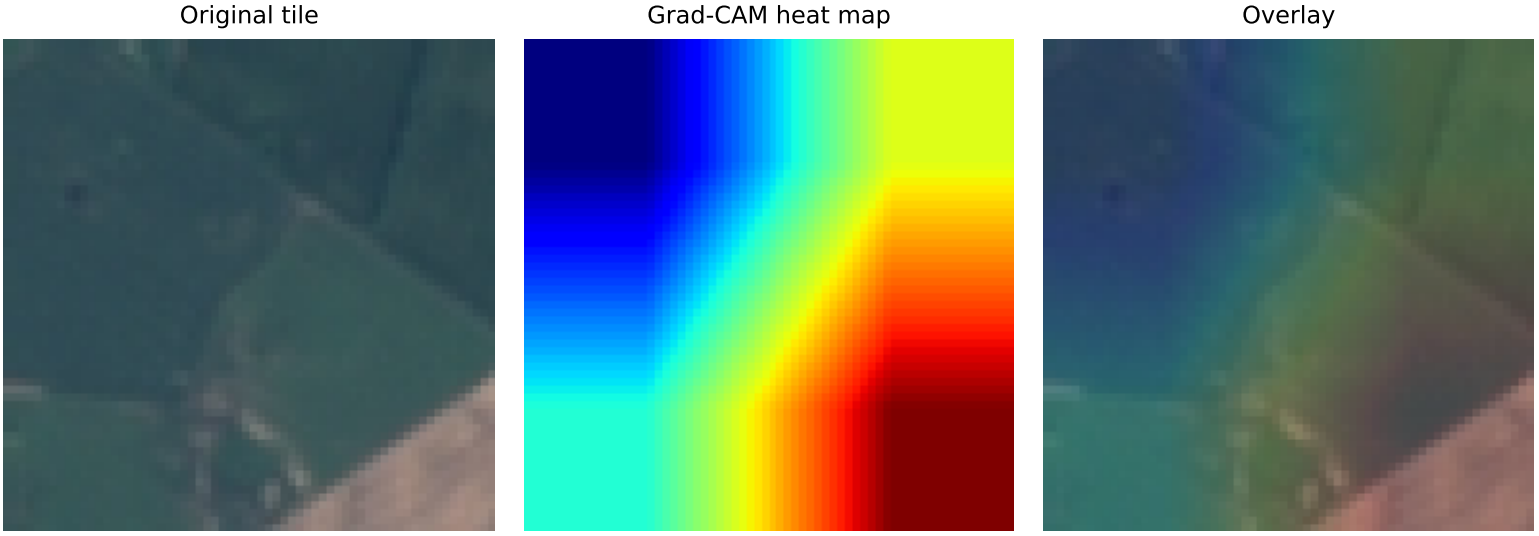


Overlay



Heat map: where the model looked. Use with care: see report section 6 (control experiment).

12. WRONG | actual: AnnualCrop | predicted: Pasture (99.7% confident)
eurosat/2750/AnnualCrop/AnnualCrop_2581.jpg



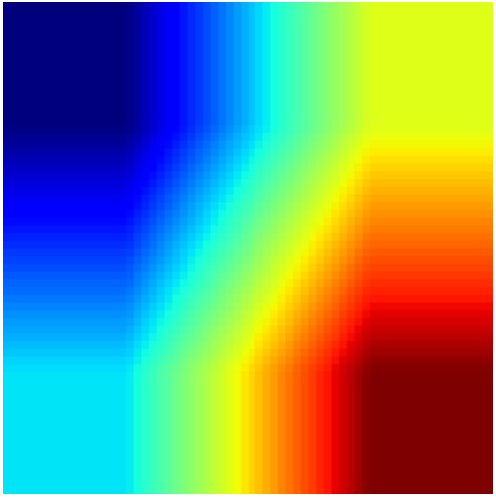
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

13. WRONG | actual: HerbaceousVegetation | predicted: Pasture (54.6% confident)
eurosat/2750/HerbaceousVegetation/HerbaceousVegetation_1308.jpg

Original tile



Grad-CAM heat map



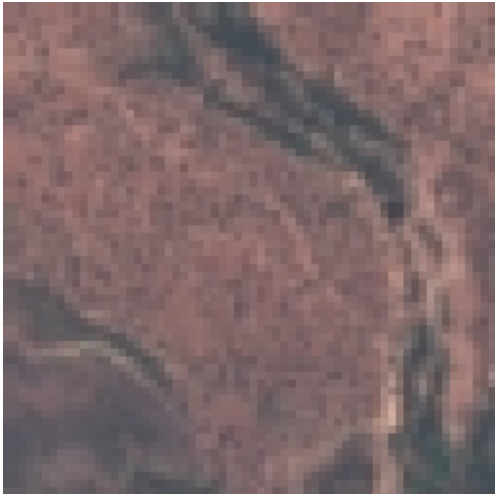
Overlay



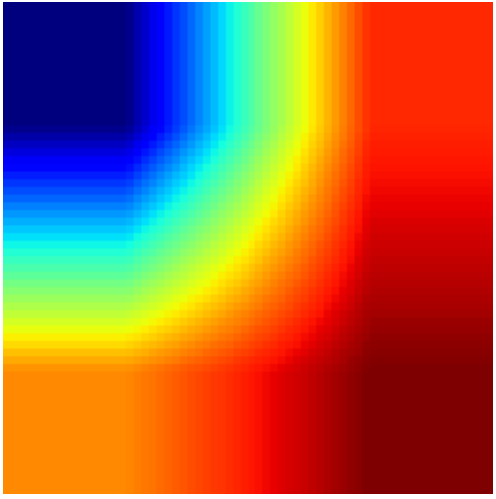
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

14. WRONG | actual: HerbaceousVegetation | predicted: PermanentCrop (80.7% confident)
eurosat/2750/HerbaceousVegetation/HerbaceousVegetation_189.jpg

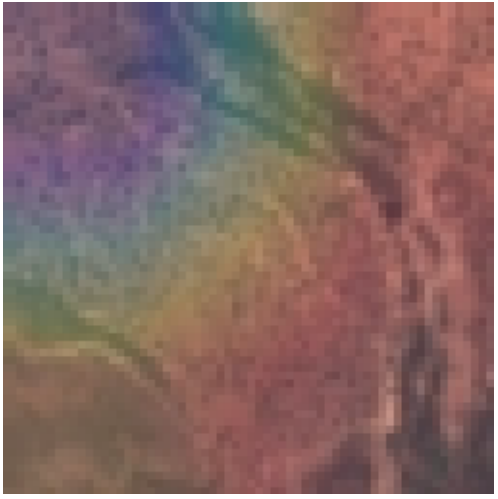
Original tile



Grad-CAM heat map



Overlay



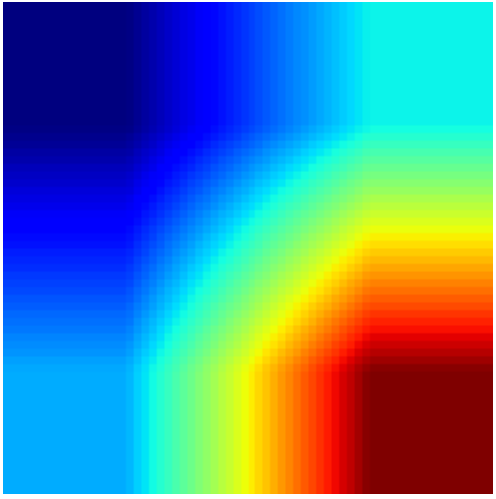
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

15. WRONG | actual: Highway | predicted: River (50.3% confident)
eurosat/2750/Highway/Highway_1654.jpg

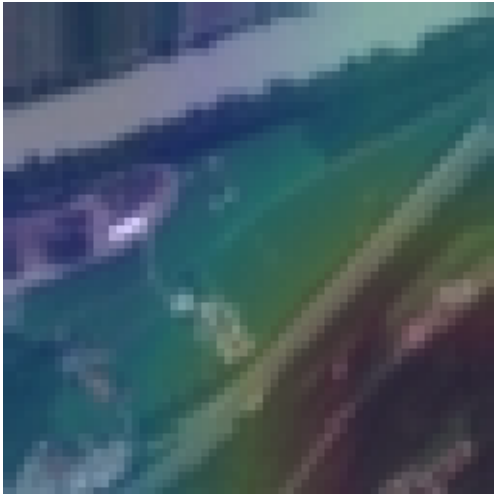
Original tile



Grad-CAM heat map



Overlay



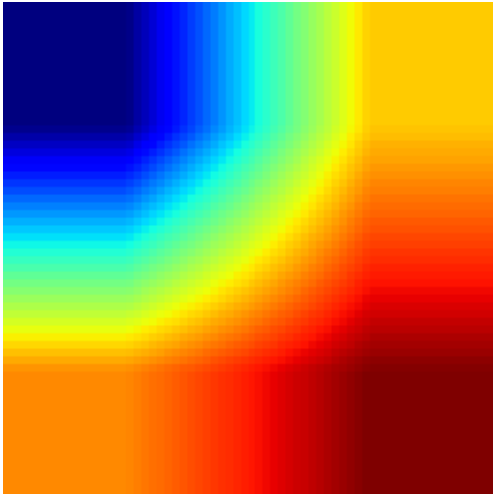
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

16. WRONG | actual: Pasture | predicted: AnnualCrop (85.8% confident)
eurosat/2750/Pasture/Pasture_1163.jpg

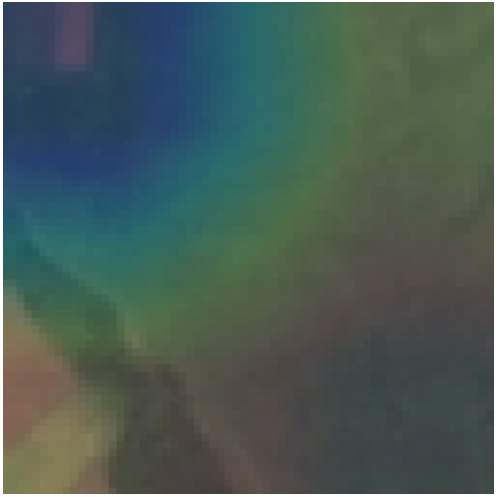
Original tile



Grad-CAM heat map



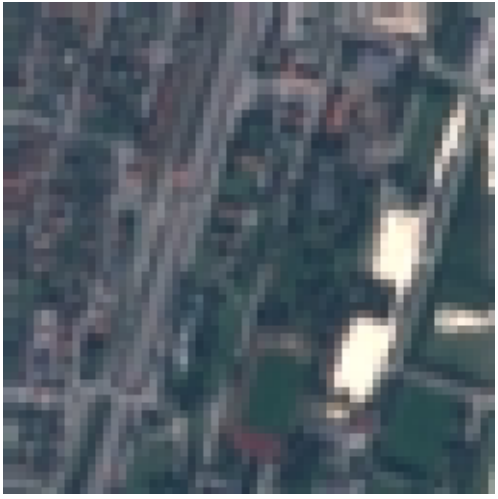
Overlay



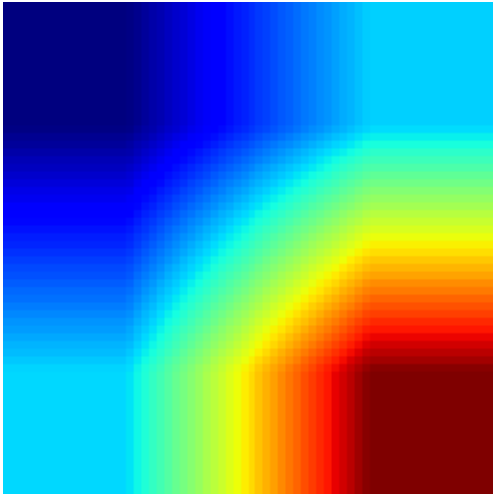
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

17. WRONG | actual: Industrial | predicted: Highway (87.0% confident)
eurosat/2750/Industrial/Industrial_1386.jpg

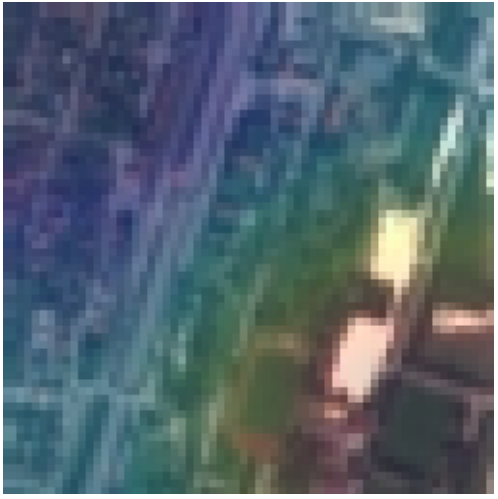
Original tile



Grad-CAM heat map



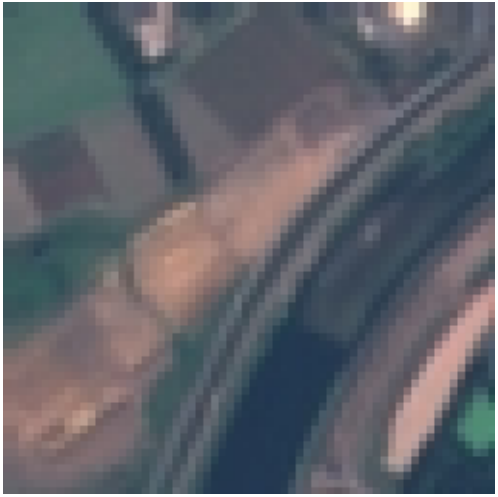
Overlay



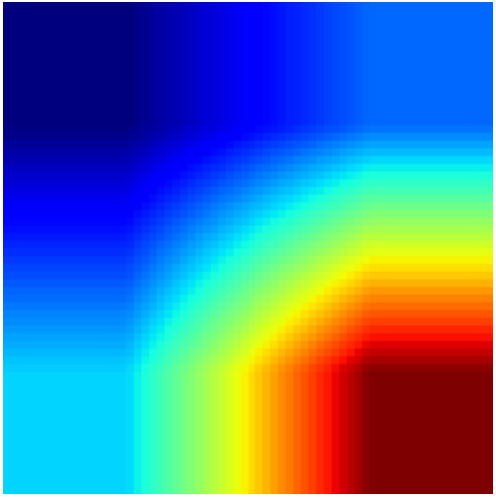
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

18. WRONG | actual: River | predicted: Highway (66.4% confident)
eurosat/2750/River/River_2320.jpg

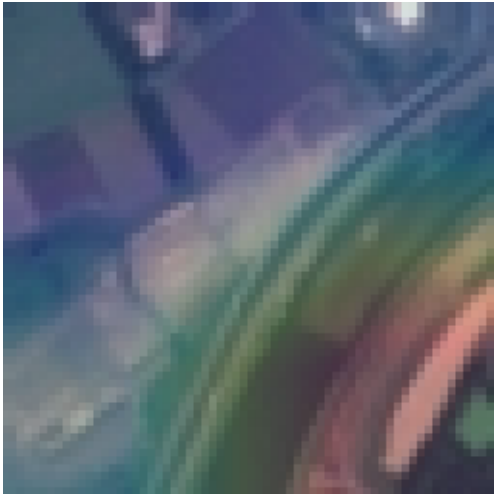
Original tile



Grad-CAM heat map



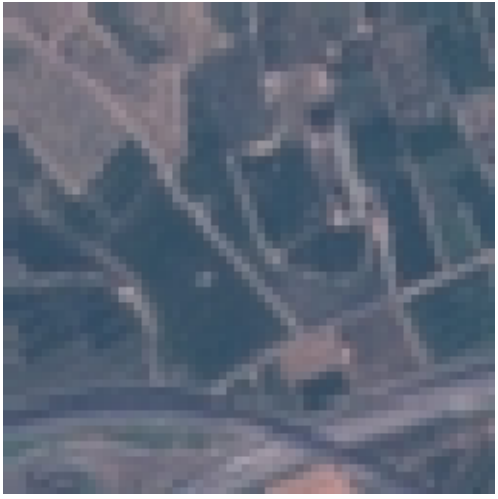
Overlay



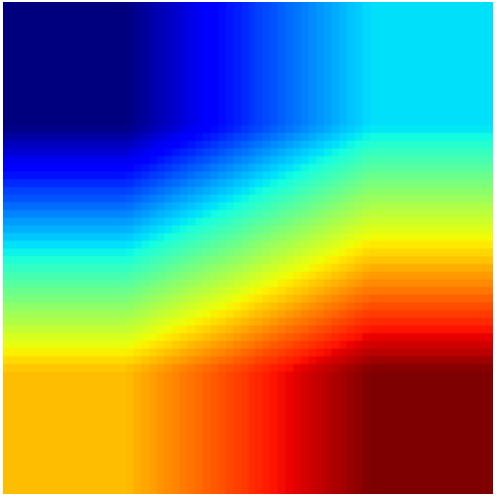
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

19. WRONG | actual: PermanentCrop | predicted: Highway (84.0% confident)
eurosat/2750/PermanentCrop/PermanentCrop_416.jpg

Original tile



Grad-CAM heat map



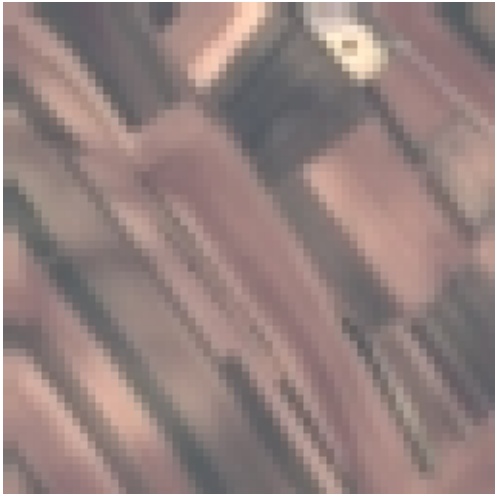
Overlay



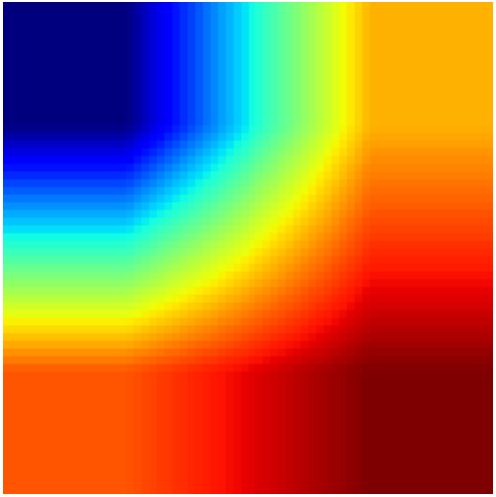
Heat map: where the model looked. Use with care: see report section 6 (control experiment).

20. WRONG | actual: PermanentCrop | predicted: AnnualCrop (93.3% confident)
eurosat/2750/PermanentCrop/PermanentCrop_2176.jpg

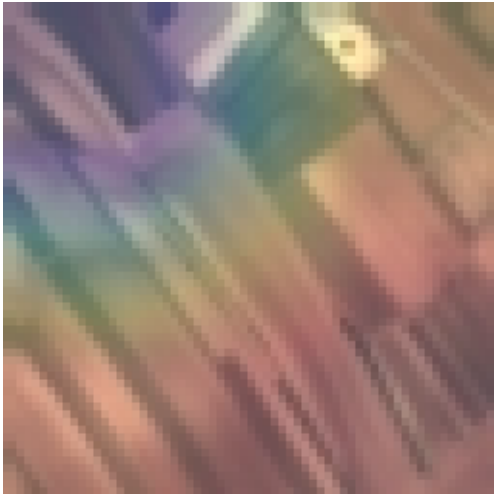
Original tile



Grad-CAM heat map



Overlay



Heat map: where the model looked. Use with care: see report section 6 (control experiment).